

DevOps Lab Program-2

Develop a multi-stage Dockerfile for container orchestration

Structure of the program:

Program-2:

→Dockerfile

→src/index.js

→package.json

→package-lock.json

→node_modules

Steps for Execution:

Step 1: Create a Dockerfile

```
GNU nano 8.1
FROM node:20-alpine AS builder

WORKDIR /app

COPY package.json package-lock.json ./
RUN npm install

COPY . .

RUN npm run build

FROM node:20-alpine

WORKDIR /app

COPY --from=builder /app/package.json ./
COPY --from=builder /app/package-lock.json ./
COPY --from=builder /app/dist ./dist
COPY --from=builder /app/node_modules ./node_modules

EXPOSE 3000

CMD ["node" ,"dist/index.js"]
```

Step 2: Create a Package.json using npm init -y

Step 3: Create a src folder and write index.js file inside it

A screenshot of a terminal window. The title bar at the top shows a window icon, a close button, and the text "1rv24mc083-rak". The terminal content shows the "GNU nano 8.1" prompt. Below the prompt, the following JavaScript code is displayed in a monospaced font with syntax highlighting:

```
const express = require('express');
const app = express();
const port = 3000;

app.get('/', (req, res) => {
  res.send("hello");
});
app.listen(port, ()=>{
  console.log(`server is running in ${port}`);
});
```

Step 4: Inside the terminal create Docker image.

Step 5: Inside the terminal run the container

```
1rv24mc083-rakshithadas@1rv24mc083:~$ docker buildx build
ERROR: docker: 'docker buildx build' requires 1 argument

Usage: docker buildx build [OPTIONS] PATH | URL | -

Run 'docker buildx build --help' for more information
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/lab2$ sudo docker build -t program2 .
[+] Building 28.5s (15/15) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 426B
=> [internal] load metadata for docker.io/library/node:20-alpine
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [builder 1/6] FROM docker.io/library/node:20-alpine@sha256:658d0f63e501824d6c23e06d4bb95c71e7d704537c9d9272f488ac03a370d448
=> => resolve docker.io/library/node:20-alpine@sha256:658d0f63e501824d6c23e06d4bb95c71e7d704537c9d9272f488ac03a370d448 7.67kB / 7.67kB
=> => sha256:658d0f63e501824d6c23e06d4bb95c71e7d704537c9d9272f488ac03a370d448 7.67kB / 7.67kB
=> => sha256:fcbb8f7d018707c656a4da2eea8a15f2893d2258093fea9ff2ea5ea1c8a82112 1.72kB / 1.72kB
=> => sha256:e80b0510ba947015cacdde3d23dc3d761e399971c074a1bf32eea7e44510524 6.52kB / 6.52kB
=> => sha256:1074353eec0db2c1d81d5af2671e56e00cf5738486f5762609ea33d606f88612 3.86MB / 3.86MB
=> => sha256:8d06ba6946d1f299d8f962e37906c77006df33d47050430a57f7893ec35af697 42.78MB / 42.78MB
=> => sha256:cb3325e64457574e24f92246e3da3376946e473d636209e1390eac47b50b26a3 1.26MB / 1.26MB
=> => extracting sha256:1074353eec0db2c1d81d5af2671e56e00cf5738486f5762609ea33d606f88612 443B / 443B
=> => extracting sha256:8d06ba6946d1f299d8f962e37906c77006df33d47050430a57f7893ec35af697 42.78MB / 42.78MB
=> => extracting sha256:cb3325e64457574e24f92246e3da3376946e473d636209e1390eac47b50b26a3 1.26MB / 1.26MB
=> => extracting sha256:fd1849a5c548bc65ee47a64498951bda5d40e87d08efe9dca69b5c8cdedc7a52 443B / 443B
=> [internal] load build context
=> => transferring context: 2.70MB
=> [builder 2/6] WORKDIR /app
=> [builder 3/6] COPY package.json package-lock.json ./
=> [builder 4/6] RUN npm install
=> [builder 5/6] COPY . .
=> [builder 6/6] RUN npm run build
=> [stage-1 3/6] COPY --from=builder /app/package.json ./
=> [stage-1 4/6] COPY --from=builder /app/package-lock.json ./
=> [stage-1 5/6] COPY --from=builder /app/dist ./dist
=> [stage-1 6/6] COPY --from=builder /app/node_modules ./node_modules
=> exporting to image
=> => exporting layers
=> => writing image sha256:6e2123227237144390e0a0cfa5f29b55ade269dec073033dcb32208fff337620
=> => naming to docker.io/library/program2
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/lab2$ sudo docker images
REPOSITORY          TAG         IMAGE ID      CREATED        SIZE
program2             latest      6e2123227237  21 seconds ago 137MB
program1             latest      2dc59dcb6250  5 minutes ago  133MB
lab6                 latest      5f211a0185ab  5 weeks ago   137MB
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/lab2$ sudo docker run -p 3000:3000 --name program2 program2
server is running in 3000
```

Step 6: Check in your browser and goto <http://localhost:3000>



hello

Step 7: Verify the containers in the terminal